

5.2.1 User Authentication Interface

The User Authentication Module provides secure access to the system by allowing authorized users to log in using their registered email and password. After successful authentication, the system verifies the user's role and grants access to the appropriate dashboard based on their responsibilities, such as Administrator, Inspector, Safety Officer, Technician, and Estate Officer. This ensures that users can only access the functions assigned to their respective roles.

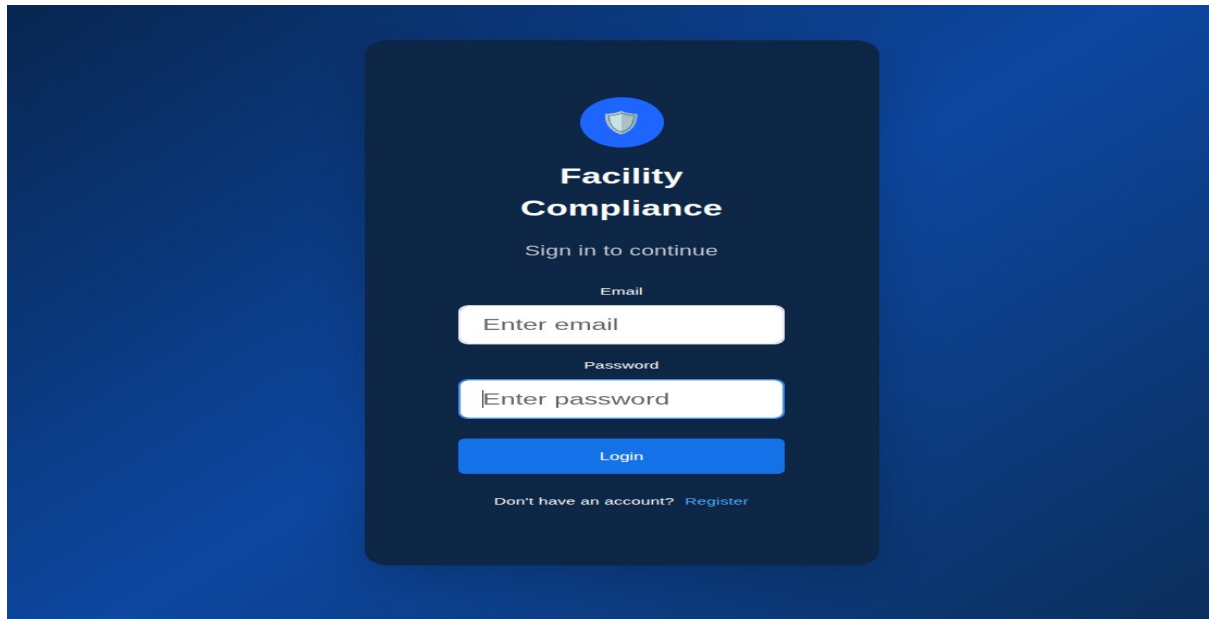


Figure 5.1: Login Page.

5.2.2 Dashboard Interface

The Dashboard Module provides users with an overview of system activities through graphical summaries and statistical information. It displays key information such as the total number of assets, scheduled inspections, compliance status, pending CAPA tasks, reported incidents, and certificates nearing expiry. The dashboard enables users to quickly monitor the overall safety and compliance performance of facility assets.

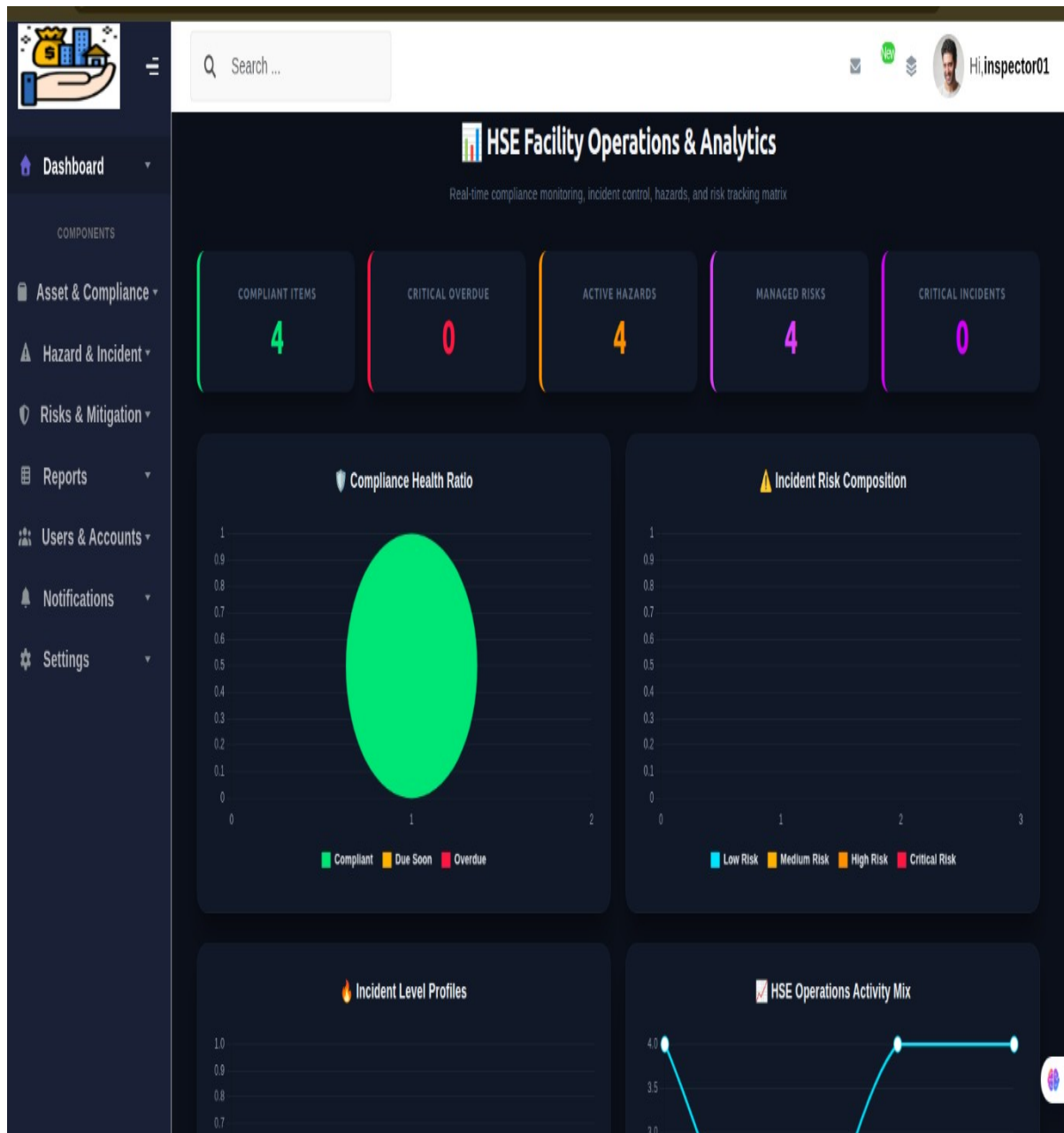


Figure 5.2: Dashboard Interface.

5.2.3 Asset Management Interface

The Asset Management Module enables users to register, update, view, and manage facility assets such as fire protection systems, lifts, electrical installations, boilers, pressure vessels, and environmental permits. Each asset is assigned unique information including its location, category, serial number, and operational status, allowing efficient monitoring throughout its lifecycle.

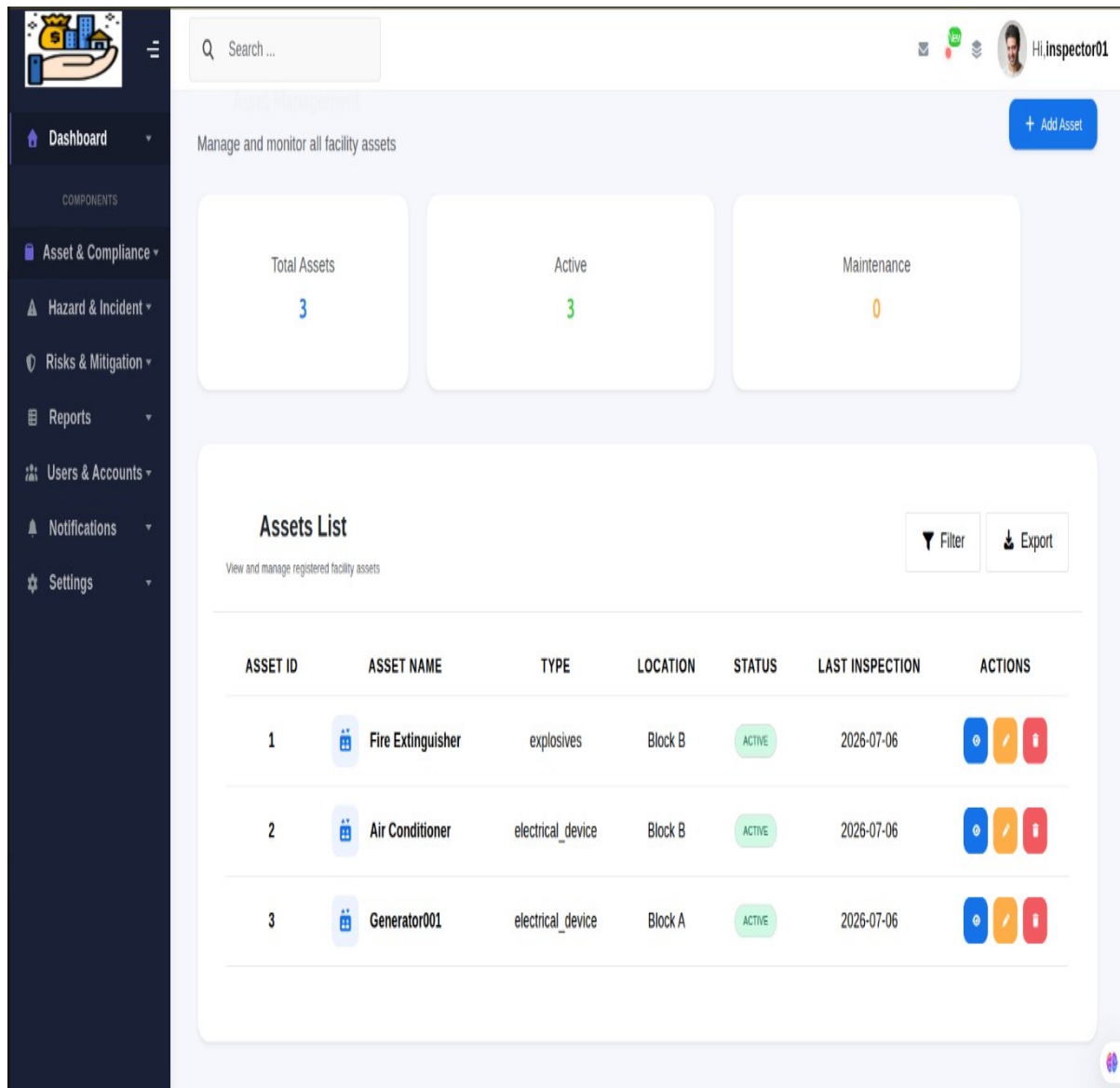


Figure 5.3: Asset Management Interface

5.2.4 Inspection Management Interface

The Inspection Management Module allows inspectors to schedule inspections, record inspection findings, upload inspection reports, and update the compliance status of facility assets. The module maintains inspection history and automatically identifies overdue inspections to support statutory compliance.

The screenshot displays the Inspection Management Interface. On the left is a dark sidebar with navigation links: Dashboard, Asset & Compliance (with sub-links for Assets, Compliance, Hazard & Incident, Hazard, and Incidents), Risks & Mitigation, Reports, Users & Accounts, Notifications, and Settings. The main content area has a top header with a search bar and a user profile 'Hi, Inspector01'. Below the header, the asset 'Fire Extinguisher' is shown with location 'Block B' and a green 'Operational' status tag. A summary section contains four cards: 'Total Compliances' (2), 'Compliant' (2), 'Due Soon' (0), and 'Over Due' (0). The 'Compliance Register' section features a table with columns: COMPLIANCE, FREQUENCY, LAST INSPECTION, NEXT DUE DATE, STATUS, and ACTIONS. It lists two entries: 'quarterly inspection' (Quarterly, 2026-07-17, 2026-07-18, COMPLIANT) and 'monthly inspection' (Monthly, 2026-07-06, 2026-07-24, COMPLIANT). Each entry has edit and delete icons in the Actions column. An '+ Add Compliance' button is located at the top right of the register section.

COMPLIANCE	FREQUENCY	LAST INSPECTION	NEXT DUE DATE	STATUS	ACTIONS
quarterly inspection	Quarterly	2026-07-17	2026-07-18	COMPLIANT	
monthly inspection	Monthly	2026-07-06	2026-07-24	COMPLIANT	

Figure 5.4: Inspection Management Interface.

5.2.5 Compliance Tracking Interface

The Compliance Tracking Module monitors the compliance status of all registered assets by evaluating inspection results, certification validity, and regulatory requirements. Assets are automatically categorized as compliant, non-compliant, pending inspection, or overdue, enabling timely corrective actions.

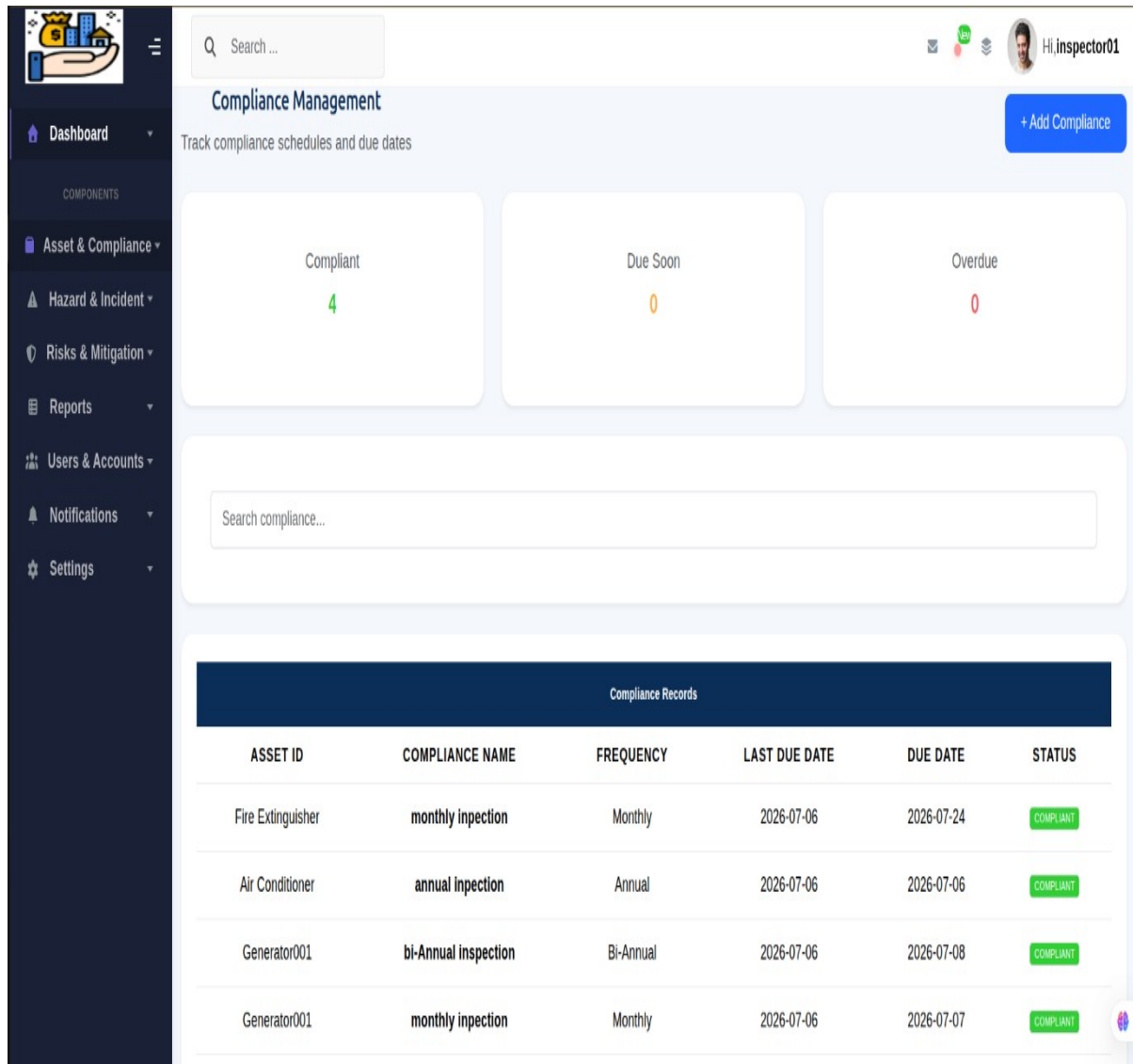


Figure 5.5: Compliance tracking interface

5.2.6 Risk Assessment Interface

The Risk Assessment Module enables Safety Officers to identify hazards, assess risk levels, assign mitigation measures, and maintain a comprehensive risk register. Risks are categorized according to their likelihood and severity to assist in prioritizing safety interventions.

The screenshot displays the Risk Assessment Interface. On the left is a dark sidebar with navigation options: Dashboard, Asset & Compliance, Hazard & Incident, Risks & Mitigation, Reports, Users & Accounts, Notifications, and Settings. The main area features a 'Risk Register' header with a '+ Add Risk' button. Below the header is a table listing identified risks and hazard assessments.

HAZARD ID	RISK TITLE	LIKELIHOOD	SEVERITY	RISK LEVEL	DESCRIPTION	STATUS	ACTIONS
HZD-001	Lift Sensor Failure	Likely	High	Critical	Failure of lift sensor may cause unsafe operation.	Active	  
HZD-002	Expired Fire Extinguisher	Possible	Medium	Moderate	Fire extinguisher certificate has expired.	Pending	  
HZD-003	Electrical Short Circuit	Almost Certain	Very High	Extreme	Damaged wiring may cause fire outbreak.	Active	  

Figure 5.6: Risk Assessment Interface.

5.2.7 Incident Management Interface

The Incident Management Module allows users to report incidents and near-miss events associated with facility assets. The system records incident details, affected assets, severity level, supporting evidence, and investigation results, facilitating effective incident management and future analysis.

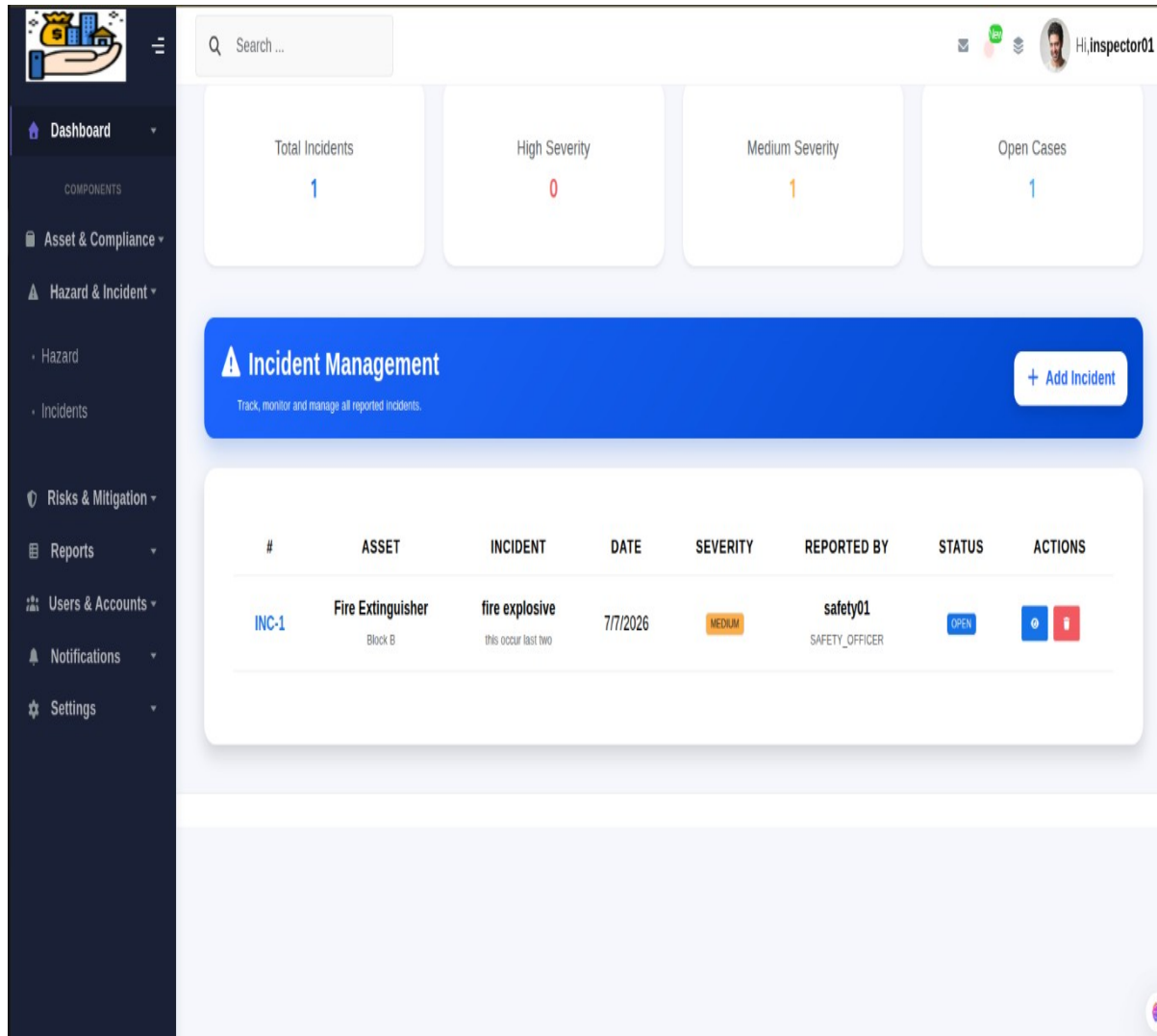


Figure 5.7: Incident Management Interface.

5.2.8 CAPA Management Interface

The Corrective and Preventive Action (CAPA) Management Module enables Safety Officers to create corrective and preventive action tasks after risks, incidents, or inspection findings are identified. CAPA tasks are assigned to technicians, monitored until completion, and verified before the asset's compliance status is updated.

The screenshot displays the CAPA Management Interface. On the left is a dark sidebar menu with icons and labels for various system components. The main content area has a light blue header with a search bar and user information. Below the header, there are three main sections:

- Dehydration:** A card showing 'Dehydration' with a 'MEDIUM' risk level, a 'POSSIBLE' status, and an 'OPEN' button.
- Mitigation Actions:** A card with a blue header and an 'Add Action' button. It contains a table with the following data:

#	ASSET	HAZARD	ACTION TITLE	ASSIGNED TO	TARGET DATE	STATUS
1	Air Conditioner ELECTRICAL_DEVICE	Temperature Shock	stop to use temporary	technician01 TECHNICIAN	09/07/2026	PENDING
- CAPA Actions (Corrective & Preventive):** A card with a red header and an 'Assign' button. It contains a table with the following data:

ACTION TITLE	ASSIGNED TO	STATUS	DUE DATE	ACTION	CREATED
sise	technician01	PENDING	2026-07-09	Approve	2026-07-06

Figure 5.8: CAPA Management Interface.

5.2.9 Reports Interface

The Reports Module generates comprehensive reports on inspections, compliance status, incidents, risk assessments, CAPA activities, and certification records. These reports assist management in monitoring system performance, supporting audits, and making informed decisions regarding facility safety.

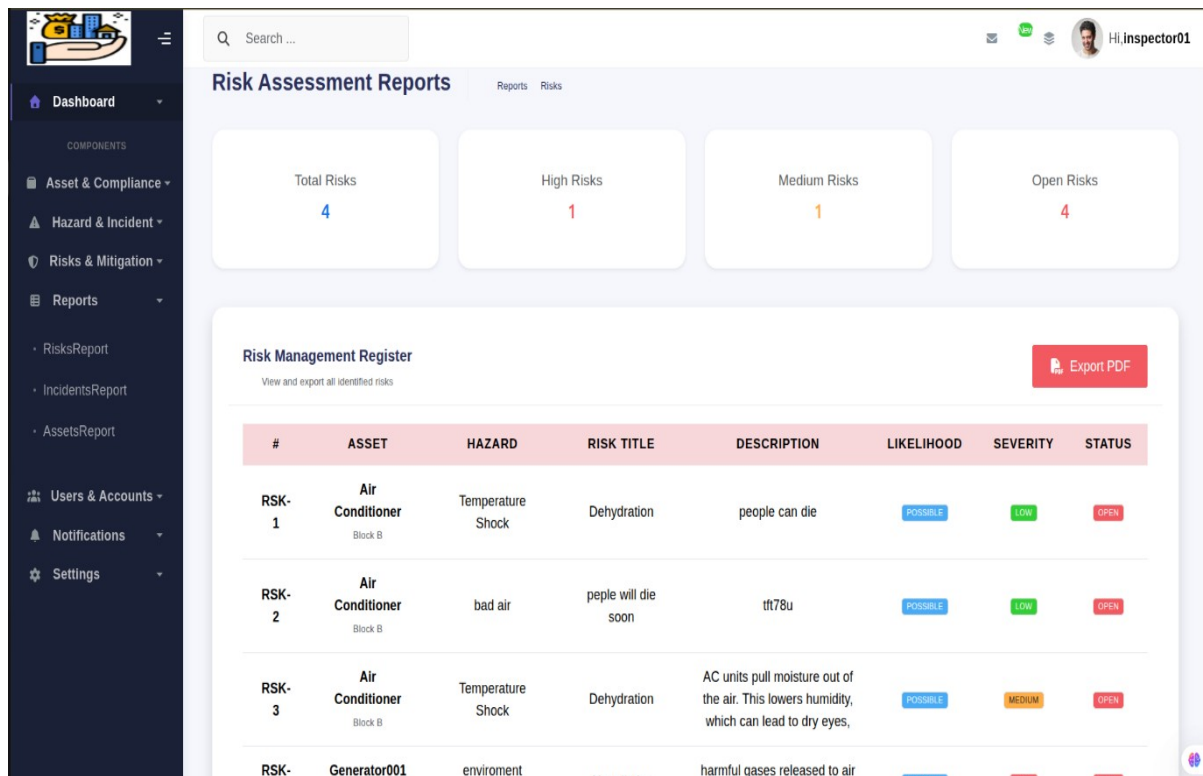


Figure 5.9: Reports Interface.